

VELLANKI NITISH BALA SOURI RAJ

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SUMMARY

Pre-final year B.Tech CSE student (CGPA: 9.11/10) with hands-on experience building **full-stack web applications** using **Node.js, Express.js, React, and PostgreSQL**. Proven ability to design REST APIs, implement JWT authentication, integrate Redis caching, and deploy via CI/CD pipelines. Ranked 61st/4,000+ teams at NEC 2025 (IIT Bombay). Seeking **Full Stack Development Intern** role to build and scale backend systems in a fast-paced SaaS environment.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java

Backend: Node.js, Express.js, NestJS (learning), FastAPI, REST APIs, GraphQL (familiar), Microservices

Frontend: React.js, Next.js (familiar), TailwindCSS, Radix UI

Databases: PostgreSQL, MongoDB, Redis — schema design, query optimisation, caching

Auth & Security: JWT, token-bucket rate limiting, Base62 encoding, stateless auth

DevOps & Tools: Docker, Docker Compose, CI/CD (Vercel, Render), Git, Postman, Drizzle ORM

AI-Assisted Coding: Cursor, Replit (Agentic AI workflows)

Certifications: Microsoft Azure AZ-204 | Lyzr AI Agent Architect

EXPERIENCE

AI & Prompt Engineering Intern

Jun 2025 – Jul 2025

VaultofCodes

Remote

- Designed and iterated 30+ prompt templates for LLM workflows; improved task-completion accuracy by ~25% using chain-of-thought structuring.
- Integrated LLM API outputs into **REST API backend pipelines**, bridging AI inference layers with production-grade server-side logic.
- Reduced average prompt token usage by ~20%, directly lowering inference and API costs.

AI & ML Intern

Apr 2025 – Jun 2025

GDG – EduSkill AP

Visakhapatnam

- Built end-to-end image classification pipeline (OpenCV + SVM) achieving ~92% validation accuracy; improved by ~8% via HOG feature engineering.
- Documented data preprocessing and model evaluation workflows, contributing to team knowledge base.

PROJECTS

Autonomous Trading Agent | *Python, LangChain, OpenRouter, CCXT, Streamlit, Pandas, Plotly, ta* | 🌐

- Developed an **autonomous cryptocurrency trading agent** using a **LangChain multi-agent framework** and OpenAI/OpenRouter APIs to automatically execute technical analysis and generate trading signals.
- Engineered a **market regime detection engine** leveraging the **ta** (Technical Analysis) library to compute RSI, MACD, Bollinger Bands, and ATR, classifying market states into Bullish, Bearish, and Sideways regimes.
- Architected a **resilient hybrid data ingestion pipeline** via **CCXT** routing queries between global (Binance) and INR spot markets (WazirX); implemented **exponential backoff auto-retry** logic to handle strict API rate limits.
- Designed an **interactive Streamlit dashboard** featuring real-time Plotly candlestick charts, system diagnostics health-checks, and downloadable analysis reports with regime-specific risk-mitigation measures.

SafeTrip AI — Smart Trip Planner & Emergency Assistant | *React, Node.js, Express.js, PostgreSQL* | 🌐

- Built a **production-grade full-stack MVP** with React 18 + TypeScript frontend and **Node.js/Express.js backend**, following a clean MVC architecture with shared TypeScript types.
- Designed and implemented **10+ REST API endpoints** covering user auth, trip CRUD, transport selection, and emergency nearby-search (`GET /api/emergency/nearby?lat=&lng=`).
- Engineered **PostgreSQL schema** (users, trips, trip_places, transport_options, safety_info, emergency_contacts) with Drizzle ORM; wrote seed scripts for development data.
- Integrated **SOS emergency feature** fetching nearby hospitals and police stations via geolocation, directly supporting life-critical workflows — highly relevant for a healthcare SaaS platform.

- Structured codebase for scalability: shared **/shared** layer (DB schema, validation schemas, route definitions) enabling type-safe API contracts between frontend and backend.

Scalable URL Shortener System | *Node.js, PostgreSQL, Redis, Docker* |

- Engineered a high-concurrency **full-stack backend** handling 1,000+ concurrent requests with sub-50ms median latency.
- Reduced PostgreSQL query load by ~60% through **Redis caching** of hot URLs (response time: ~120ms → ~45ms), demonstrating production-grade **database + caching integration**.
- Implemented **token-bucket rate limiting** and Base62 encoding (3.5 trillion collision-free codes), maintaining 99%+ uptime under stress tests.
- Containerised full stack with **Docker Compose** enabling one-command local and production deployment.

Skill Connect — Full Stack Web Platform | *React.js, Node.js/FastAPI, PostgreSQL* |

- Built a **full-stack web application** with React.js frontend and backend serving real-time REST API interactions.
- Implemented stateless **JWT authentication** with refresh-token rotation, securing all protected endpoints — mirroring enterprise auth patterns.
- Optimised **PostgreSQL queries** via indexed lookups, improving response time by ~40% under load.
- Deployed via **CI/CD pipelines** on Vercel (frontend) and Render (backend), achieving zero-downtime releases.

AI Resume Analyzer & Job Matcher | *FastAPI, NLP, scikit-learn* |

- Integrated ML inference (TF-IDF + cosine similarity) into a **FastAPI backend** processing 50+ resumes/minute; achieved ~85% matching accuracy.
- Designed semantic search pipeline reducing manual screening time by ~70%; reduced false positives by ~30% via threshold tuning.

EDUCATION

Vignan's Institute of Information Technology

B.Tech in Computer Science & Engineering | **CGPA: 9.11 / 10**

Visakhapatnam, India

2023 – 2027

ACHIEVEMENTS & LEADERSHIP

- Ranked **61 / 4,000+** teams — NEC 2025, IIT Bombay (National-level engineering competition).
- CodeChef **2★** rated | Strong foundations in Arrays, Hashing, Recursion, and Dynamic Programming.
- **Web Development Lead** — EDCELL VIIT | Led all web initiatives for the college entrepreneurship cell.
- Experience with **AI-assisted coding tools**: Cursor, Replit, Windsurf — accelerates development velocity in fast-paced SaaS sprints.